A Finite-Rate Selection Law for Annealed Gibbs Sampling

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Abstract

A sampler can generate plausible examples in the wrong proportions. We show how this happens in annealed Gibbs sampling, which updates one variable at a time while lowering the temperature. In the Curie–Weiss model, binary variables tend to share one sign, giving two modes, and a weak bias, the field, favors one mode. Each run selects a mode as the modes appear, when the noise of the updates can still outweigh the field, and rarely switches afterwards. The samples therefore keep only a fraction of the model's log-odds between the modes. We turn the classical probability of this selection, due to Kondepudi and Nelson, into a law for this fraction. The law agrees closely with an exact calculation of the sampler and explains why slower cooling helps little: on a linear ramp, cooling sixteen times more slowly only doubles the fraction. In a Hopfield model, competition among stored patterns raises the fraction. On one linear ramp, we predict the fraction as the exact Curie–Weiss fraction times a competition factor derived under a Gaussian approximation. This pre-registered prediction is within 3% of the measured fraction for three patterns at three sizes; for four patterns, the measurement is 6% lower.[1]

1. Introduction

Suppose a model of handwritten digits assigns twice as much probability to "3" as to "8". A sampler should then produce about twice as many "3"s as "8"s over many runs. It can produce recognizable examples of both digits and still get this proportion wrong, and Monte Carlo estimates from its samples are then biased. How does the sampler change these proportions?

We study annealed Gibbs sampling, which redraws one variable at a time from its conditional distribution. At high temperature the model's distribution is broad; at low temperature its probability concentrates in modes, regions of likely outcomes separated by unlikely ones. As the temperature falls, the regions between modes become even less likely and the sampler stops mixing between them: later updates mostly sharpen the mode each run selected and rarely undo the selection [9].

The sampler can therefore get the mode proportions wrong even when the model is right. To measure this error alone, we use the Curie–Weiss (CW) model, whose probabilities, and its Gibbs sampler's, can be computed exactly. Its binary variables tend to align, giving two modes below a critical temperature: most variables positive, or most negative. A weak field, a bias on every variable, favors the positive mode, but each run selects its mode near the critical temperature, where the noise of the Gibbs updates can outweigh the field.

1. Code and details: https://github.com/ali-izhar/finite-rate-selection-law

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We turn the classical probability of selecting the positive mode, due to Kondepudi and Nelson [7] [see also 6, 10], into a *selection law* for the fraction of the model's log-odds between the modes that the samples keep, and show why slower cooling helps little. Extended to a Hopfield model, whose modes are stored patterns and their negatives, the argument predicts that competition among patterns multiplies the exact CW fraction by a factor that grows with their number. A pre-registered test on one linear cooling ramp agrees to within 3% for three patterns at three sizes; for four patterns, the measurement is 6% lower than predicted.

2. An exact model of mode selection

The model's state, or *configuration*, is a vector $x = (x_1, \ldots, x_N)$ of N binary variables $x_i \in \{-1, +1\}$, called spins. The CW model gives higher probability to configurations whose spins agree:

$$\pi_{\beta,G}(x) \propto \exp\left[\frac{\beta}{2(N-1)} \sum_{i \neq j} x_i x_j + \beta G \sum_i x_i\right]. \tag{1}$$

The *inverse temperature* β sets how strongly spins prefer to agree: at $\beta = 0$ all configurations are equally likely, and increasing β, which is cooling, concentrates probability on aligned configurations. The field G is fixed during a run and favors positive spins when $G > 0$. At fixed β, the Gibbs sampler converges to the *equilibrium distribution* $\pi_{\beta,G}$. The *magnetization* $m = N^{-1} \sum_i x_i$ summarizes a configuration: $m > 0$ means mostly positive spins, $m < 0$ mostly negative. Without a field, as $N \to \infty$, the equilibrium m concentrates at 0 for $\beta < 1$ and at two opposite nonzero values for $\beta > 1$. Configurations with m near either value form the two modes, and $\beta = 1$ marks the *transition*, at the critical temperature.

A run starts at some $\beta < 1$, from the equilibrium distribution at that or a lower β. Each Gibbs update redraws one uniformly chosen spin; updates occur at random times, N per unit time on average, and this unit is an *expected sweep*. A cooling schedule raises β past the transition to a final value, by which the run has settled into one mode. At a given β, configurations with the same number k of positive spins are equally probable in equilibrium and change k at the same rates, which lets k, and so $m = 2k/N - 1$, alone describe the run: we track $N + 1$ probabilities, one for each $k = 0, \ldots, N$, instead of 2^N, one for each configuration [8]. We compute them exactly, without sampling error, by numerically solving the linear equations that govern their change (Appendix A).

Each run's final configuration is its sample. Let q_{chain} be the probability that a sample has $m > 0$, and q_{eq} the probability of $m > 0$ under the equilibrium distribution at the final β, both conditioned on $m \neq 0$. The fraction of the model's log-odds, $\operatorname{logit} q_{\text{eq}}$, that the samples keep is

$$c = \frac{\operatorname{logit} q_{\text{chain}}}{\operatorname{logit} q_{\text{eq}}}, \qquad \operatorname{logit} q = \log \frac{q}{1-q}, \tag{2}$$

for $q_{\text{eq}} \neq 1/2$: $c = 1$ reproduces the equilibrium proportions, and $c = 0$ gives equal mode probabilities. If equilibrium odds are 4:1, $c = 0.2$ gives sampled odds of $4^{0.2}$:1 $\approx$ 1.32:1.

3. The selection law

Each run selects its mode in a short window near $\beta = 1$. There, a small random imbalance between positive and negative spins grows into one mode (Figure 1a); afterwards, switching between modes

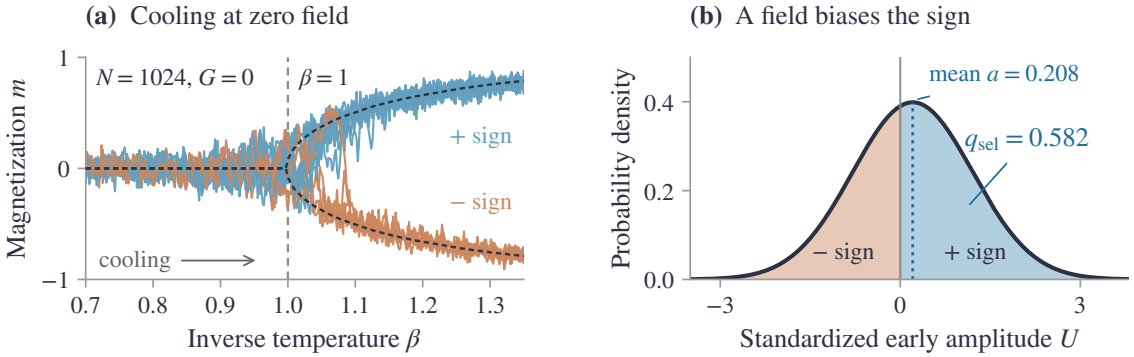


Figure 1: (a) Fourteen paths ($N = 1024$, $G = 0$) of m under the drift $-m+\tanh(\beta m)$ plus Gaussian noise, approximating the Gibbs chain, split into two modes; the dashed curves mark the most probable equilibrium magnetization. (b) Analytical density of the standardized early amplitude $U \sim \mathcal{N}(a, 1)$ on a linear ramp ($N = 2048$, $\gamma = 0.001$, $G = 0.0006125$).

takes far longer than settling within one [5, 8]. The final proportions are therefore decided in this window, the only time the field can bias them.

Near the transition, two effects move m (Appendix A). The *drift*, the mean rate of change of m, is $-m + \tanh(\beta m + \beta G)$, approximately $(\beta - 1)m + G$ near $m = 0$ and $\beta = 1$. Below $\beta = 1$ the term $(\beta - 1)m$ points against m and pulls an imbalance back toward zero; above $\beta = 1$ it points along m and amplifies the imbalance. The *noise* comes from the randomness of the Gibbs updates and adds a variance of about $2/N$ per expected sweep. On a linear ramp at cooling rate $\gamma > 0$, $\beta(t) = 1 + \gamma t$ crosses $\beta = 1$ at $t = 0$. The growth rate $\beta - 1 = \gamma t$ then rises with time, and the drift multiplies the imbalance by $e^{\gamma t^2/2}$, a factor that stays near one only for $|t|$ below a time T of order $\gamma^{-1/2}$: this is the window. Within it, the field adds an imbalance of order GT and the noise one of order $\sqrt{T/N}$ [3]. Their ratio, of order $G\sqrt{N}\,\gamma^{-1/4}$, decides whether the field or the noise sets the sign: slower cooling widens the window and favors the field, but only as $\gamma^{-1/4}$.

To turn the field-to-noise ratio into the probability that a run ends in the positive mode, we divide m by its growth factor $e^{\gamma t^2/2}$. The result, the early *amplitude*, has the sign of m and stops changing once the window closes, because later kicks are divided by a large growth factor. It is the field's push, the same in every run, plus the accumulated noise, a sum of independent Gaussian kicks; in the linear approximation it is therefore Gaussian across runs, with mean set by the field and spread by the noise. In units of its spread, it follows $U \sim \mathcal{N}(a, 1)$, where a is this field-to-noise ratio with its exact prefactor. In the linear approximation, a run ends with $m > 0$ when $U > 0$, which happens with probability $\Phi(a)$ (the blue area in Figure 1b), where Φ is the standard Gaussian CDF. For the parameters of Figure 1b, $a = 0.208$: the noise is about five times stronger than the field, yet 58% of runs end in the favored mode. This is the classical result [6, 7, Appendix A]

$$q_{\text{sel}} \simeq \Phi(a), \qquad a = G\sqrt{N}(\pi/\gamma)^{1/4}, \tag{3}$$

with q_{sel} the probability of selecting $m > 0$. Without later switching, $q_{\text{chain}} = q_{\text{sel}}$.

With q_{sel} for q_{chain}, Eq. (2) gives the *selection law*, $c_{\text{sel}} = \operatorname{logit}\Phi(a)/\operatorname{logit} q_{\text{eq}}$. For a weak field ($|a| \ll 1$), the sampled log-odds are $\operatorname{logit}\Phi(a) \approx 4a/\sqrt{2\pi}$, proportional to a, hence to $G\sqrt{N}\,\gamma^{-1/4}$.

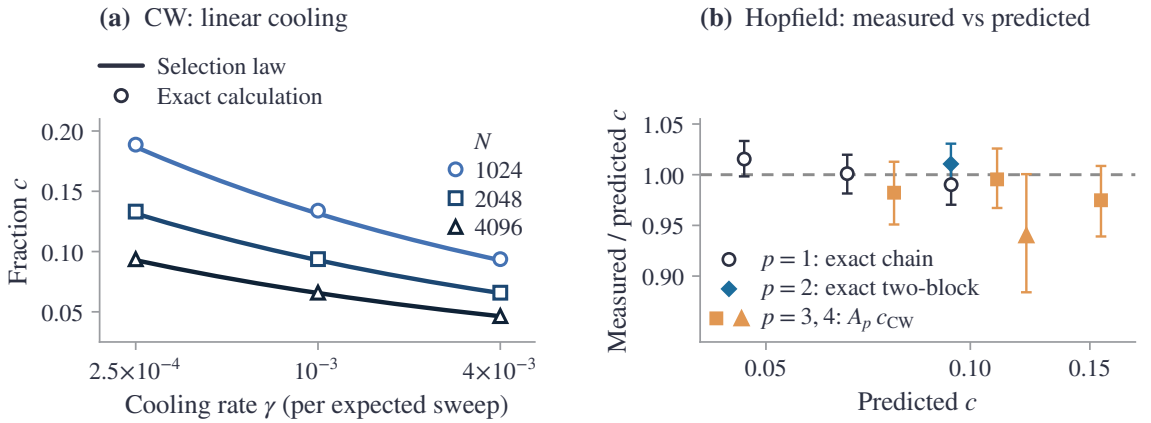


Figure 2: (a) CW ramps at fixed NG: the law is within 2%. (b) Hopfield: ratio of measured to predicted c for the eight rows of Table 1, with the predictions of Section 4; bars are 95% intervals.

The equilibrium log-odds at the final temperature grow faster with N, as NG, than the sampled log-odds: reversing all N spins leaves the coupling term unchanged and reverses the field term, βGNm, of order NG in either mode. In their ratio, the field cancels, and c measures the sampler, not the field (prefactor: Appendix A):

$$c \propto (N^2\gamma)^{-1/4}. \tag{4}$$

For a weak field, cooling sixteen times more slowly doubles c; a fourfold larger N halves it.

The selection law rests on four assumptions (Appendix A): a slow ramp ($\gamma \ll 1$), a start well below $\beta = 1$, a nearly linear drift, and little switching after selection. For bounded a, $N\gamma \gg 1$ guarantees the third [3]: the noise gathered in the window is then a fraction $(N\gamma)^{-1/2}$ of the magnitude $|m| \sim \gamma^{1/4}$ at which the drift turns nonlinear. Very slow cooling violates the last, since long enough sampling restores the equilibrium proportions; but at $N = 2048$ and $\beta = 1.1$, a zero-field switch between modes takes on average about 5×10^7 expected sweeps (the exact chain's mean passage time); our ramps last at most 3600 expected sweeps.

Exact check. On linear ramps ($N = 1024$–4096, $\gamma = 0.00025$–0.004, $NG = 1.2544$), exact c exceeds c_{sel} by 0.23–1.81% (Figure 2a; Appendix Table 2), even at $N\gamma = 0.256$, where $N\gamma \gg 1$ fails.

4. When several modes compete

CW has two modes. To ask how competition among several modes changes the fraction c, we use the Hopfield model: with $p \ll N$ stored patterns $\xi^\mu \in \{-1, +1\}^N$ of independent equiprobable entries, its modes at low temperature are the patterns and their negatives [1]. A configuration x has probability proportional to $e^{-\beta E_G(x)}$, with *energy*

$$E_G(x) = -\frac{1}{2(N-1)} \sum_{i \neq j} \sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu x_i x_j - G \sum_i \xi_i^1 x_i. \tag{5}$$

Table 1: Pre-registered Hopfield test on one linear ramp ($\gamma = 0.004$, $\beta = 0.7$ to 1.6). Predictions are the exact chain for $p = 1$, the exact two-block value for $p = 2$, and $A_p c_{\mathrm{CW}}$ for $p \geq 3$. Ratio: measured over predicted c. Intervals are 95% and resample the 50 pattern sets.

p	N	prediction	predicted c	measured c	ratio
1	1024	exact chain	0.0937	0.0928 [0.0909, 0.0947]	0.990
1	2048	exact chain	0.0659	0.0659 [0.0647, 0.0672]	1.001
1	4096	exact chain	0.0464	0.0472 [0.0464, 0.0480]	1.015
2	2048	exact two-block	0.0936	0.0946 [0.0928, 0.0965]	1.011
3	1024	$A_3 c_{\mathrm{CW}}$	0.1558	0.1518 [0.1463, 0.1571]	0.975
3	2048	$A_3 c_{\mathrm{CW}}$	0.1095	0.1090 [0.1059, 0.1123]	0.995
3	4096	$A_3 c_{\mathrm{CW}}$	0.0772	0.0758 [0.0734, 0.0782]	0.982
4	2048	$A_4 c_{\mathrm{CW}}$	0.1209	0.1137 [0.1069, 0.1210]	0.940

The field G favors pattern 1 over its negative. The overlap $m^\mu = N^{-1} \sum_i \xi_i^\mu x_i$ measures how closely a configuration resembles pattern μ: it is $+1$ at pattern μ and -1 at its negative. For $p = 1$, the variables $y_i = \xi_i^1 x_i$ turn $e^{-\beta E_G}$ into Eq. (1) and m^1 into the magnetization of y: the CW model is the one-pattern Hopfield model. Because the map is one to one and acts spin by spin, it also turns each Gibbs update of x into one of y, so a one-pattern run is a CW run (Appendix B). Each sample is labeled by the pattern with the largest $|m^\mu|$ and the sign of that overlap. Here q_{chain} and q_{eq} are the probabilities that a pattern-1 sample has the positive sign, so c compares log-odds of that sign, not pattern frequencies.

Competition raises the fraction. We assume that the overlaps' early amplitudes (Section 3) are independent Gaussians of common variance; that the field shifts only the mean of pattern 1's; and that a run ends in the pattern whose early amplitude is largest in magnitude. Pattern 1's sign log-odds at amplitude magnitude r then equal the log ratio of a shifted Gaussian's densities at $+r$ and $-r$, which is linear in r. Winning depends only on magnitude and leaves these log-odds unchanged at each r, but the winner of p amplitudes has a larger average magnitude than a single one; to first order in the field, its sign log-odds are multiplied by the ratio of the two average magnitudes:

$$A_p = \frac{\mathbb{E} \max_{1 \leq \mu \leq p} |Z_\mu|}{\mathbb{E}|Z|}, \qquad c_{\mathrm{pred}} = A_p c_{\mathrm{CW}}, \tag{6}$$

where Z and the Z_μ are independent standard Gaussians: $|Z_\mu|$ is the size of pattern μ's early amplitude in units of its spread; the denominator is the average size of a single amplitude, as in CW, where the one pattern always wins. Here c_{CW} is the exact CW fraction on the test's ramp (below), not the selection law's value, so the test below checks Eq. (6), not the selection law. Appendix C derives Eq. (6); numerical integration gives $A_1 = 1$, $A_2 = \sqrt{2}$, $A_3 = 1.662$, and $A_4 = 1.836$.

For two patterns, the prediction needs no Gaussian assumption. In the variables $y_i = \xi_i^1 x_i$, the spins where the two patterns agree and those where they differ form two CW models that do not interact, and pattern 1 wins exactly when their magnetizations share a sign (Appendix C). The exact chain of each block then gives c_{pred} for each pattern set; $A_2 c_{\mathrm{CW}}$ lies 0.5% below it.

Pre-registered test. All runs cool on the fastest ramp of the exact check, where $N\gamma$ is largest, from $\beta = 0.7$ to 1.6 at $\gamma = 0.004$. Runs start from independent unbiased spins, the equilibrium at

$\beta = 0$, and each update redraws one uniformly chosen spin, as in Section 2. Half the runs have field $+|G|$, half $-|G|$, with $NG = 1.2544$, and each row uses 50 pattern sets, independent draws of the p patterns. The measured c is the log-odds that a pattern-1 sample takes the field's sign, pooled over sets and signs, divided by the exact CW $\operatorname{logit} q_{\mathrm{eq}}$; 95% intervals resample pattern sets. This denominator cancels in the ratio of measured to predicted c; it equals the Hopfield $\operatorname{logit} q_{\mathrm{eq}}$ at $p = 1$ and is within 1.3% of it where we computed both ($p = 2$ at $N = 2048$; $p = 3$ at $N \le 1024$). We fixed the predictions, run counts and readings of the outcome in advance (Appendix B).

Result. The one-pattern runs reproduce the exact chain at three sizes (ratios 0.990–1.015), which checks the sampler and the estimator, and the two-pattern runs match the exact two-block value to within 1.1% (Table 1; Figure 2b). For three patterns at $N = 2048$, measured c is 0.995 times $A_3 c_{\mathrm{CW}}$ (95% interval $[0.967, 1.026]$); at $N = 1024$ and 4096 the ratios are 0.975 and 0.982, and their intervals also contain 1. Over these sizes, three-pattern c falls as $N^{-0.50}$ (exponent interval $[0.47, 0.53]$), as the exact CW fraction does and as Eq. (4) predicts at a fixed rate.

5. Where the competition prediction falls short

For four patterns, measured c is 6% below $A_4 c_{\mathrm{CW}}$; the interval of their ratio, $[0.884, 1.001]$, contains 1, but only just. All four $A_p c_{\mathrm{CW}}$ rows lie below their predictions, by 0.5–6%. Equation (6) assumes independent amplitudes of common variance, but random patterns are correlated: two of them agree on a fraction of spins that differs from one half by order $N^{-1/2}$. Overlap components along the eigenvectors of the pattern correlation matrix then begin to grow at slightly different β, and this spread is small next to the selection window only when $N\gamma \gg p$ (Appendix C). Here $p/(N\gamma)$ ranges from 0.18 to 0.73, so no row has $N\gamma \gg p$; a linearized Gaussian model of these components, which we did not validate, gives deficits of 0.5–3%. The deficits do not follow $p/(N\gamma)$ closely: the largest is at $p = 4$, $N = 2048$, not at $p = 3$, $N = 1024$, where $p/(N\gamma)$ is largest. These differences lie within the intervals, so the rows neither support nor rule out this explanation (Appendix B).

The test uses one cooling rate and one value of NG, in the weak-field regime where log-odds are linear in the field, so it does not show how competition depends on the cooling rate. The selection law and Eq. (6) describe single-spin Gibbs sampling; other samplers need their own calculation.

6. Discussion

The selection probability is classical [6, 7]; we turn it into the selection law for c, check the law against the exact CW calculation, and extend it to competing modes. Cooling studies of spin chains [11] measure misaligned regions and excess energy, not mode probabilities.

Annealed importance sampling weights annealed runs to estimate equilibrium averages [9]; unweighted runs keep only a fraction c. Comparing them at equal cost needs the weighted estimator's bias and variance. Diffusion models change their distribution gradually, but with an exact score they reverse a noising process without equilibrating at each step [4]; extending our calculation needs their reverse dynamics and time discretization. A variance-preserving noise schedule can miss unequal mode probabilities, while suitable schedules recover them in Gaussian mixtures and CW [2].

The exact CW calculation gives, without sampling, the mode-probability error of any cooling schedule; applying c to other models requires their equilibrium log-odds. Repeating the four-pattern test at several cooling rates would test the correlation argument, which expects the deficit to shrink as $N\gamma/p$ grows.

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Appendix A. Selection law and exact-chain calculations

This appendix derives the selection law of Section 3, states when it holds, and describes the exact calculation that tests it.

From spin updates to drift and noise. The drift and noise of Section 3 follow from single-spin updates. For a chosen spin, let $\ell_i = \beta(N-1)^{-1}\sum_{j\neq i} x_j + h$, where $h = \beta G$. Holding the other spins fixed, Eq. (1) assigns weights e^{ℓ_i} and $e^{-\ell_i}$ to its two values, so the redrawn spin has mean $\tanh \ell_i$. For large N, the neighbors' average is approximately m. A uniformly chosen spin has old mean m, so one update changes m by $[\tanh(\beta m + h) - m]/N$ in expectation. With N updates per expected sweep, the drift is $\dot m = -m + \tanh(\beta m + h)$. Near $m = 0$ and in a weak field, an update flips a spin with probability about one half, and each flip changes m by $2/N$. The noise variance per expected sweep is therefore approximately $N \times \frac{1}{2} \times (2/N)^2 = 2/N$. A short interval contains many updates that change m little; we approximate the centered sum of their jumps by a Gaussian. For small m and G near $\beta = 1$, we expand $\tanh z = z - z^3/3 + \cdots$, keep the field, linear, and cubic terms, and write G for $h = \beta G$ (equal at $\beta = 1$). This gives

$$dm = \big[(\beta - 1)m - m^3/3 + G\big]\,dt + \sqrt{2/N}\,dW_t, \tag{7}$$

where W_t is a standard Brownian motion. At $G = 0$ and with β ramped through 1, Eq. (7) is the dynamic pitchfork bifurcation with additive noise analyzed by Berglund and Gentz [3]: the drift's single stable value, $m = 0$, splits into two. The paths of Figure 1a integrate the drift $-m + \tanh(\beta m)$ with noise variance $2/N$ per expected sweep by Euler steps of 0.02, from $m = 0$, holding each of 360 equally spaced values of β from 0.7 to 1.35 for 6 sweeps (NumPy seed 4).

Removing growth without changing the sign. On a linear ramp, $\beta - 1 = \gamma t$. While m is small, omit the cubic term. Without field or noise, $\dot m = \gamma t m$ has the positive growth factor $e^{\gamma t^2/2}$. Dividing it out gives $Y_t = e^{-\gamma t^2/2} m_t$, which keeps the sign of m_t; its limit is the early amplitude of Section 3. Start at $t_0 < 0$. The product rule gives

$$Y_t = e^{-\gamma t_0^2/2} m_{t_0} + G\int_{t_0}^{t} e^{-\gamma u^2/2}\,du + \sqrt{2/N}\int_{t_0}^{t} e^{-\gamma u^2/2}\,dW_u.$$

The three terms are the remaining initial imbalance, the accumulated field, and the accumulated noise. With $w(u) = e^{-\gamma u^2/2}$, the noise integral is the limit of sums $\sum_j w(u_j)\sqrt{\Delta u}\,\epsilon_j$ of independent standard Gaussians ϵ_j: a Gaussian with mean zero and variance $\sum_j w(u_j)^2 \Delta u$.

The weights are appreciable over times of order $\gamma^{-1/2}$ around the transition. A start well below $\beta = 1$ ($|t_0| \gg \gamma^{-1/2}$) makes the factor $e^{-\gamma t_0^2/2}$ of the initial term negligible; a late end lets both integrals settle. Extending their limits to the whole real line gives the early amplitude Y with

$$\mu_Y = \mathbb{E}Y = G\int_{-\infty}^{\infty} e^{-\gamma u^2/2}\,du = G\sqrt{2\pi/\gamma},$$
$$V_Y = \mathrm{Var}(Y) = \frac{2}{N}\int_{-\infty}^{\infty} e^{-\gamma u^2}\,du = \frac{2}{N}\sqrt{\pi/\gamma}.$$

Hence $U = Y/\sqrt{V_Y} = a + Z$, where $Z \sim \mathcal{N}(0,1)$ and $a = \mu_Y/\sqrt{V_Y} = G\sqrt{N}(\pi/\gamma)^{1/4}$. The positive mode is selected when $U > 0$, with probability $\Pr(Z > -a) = \Phi(a)$, as in Eq. (3). These infinite limits approximate a finite cooling interval; the integrals must settle before the cubic term substantially changes the amplitude.

From a sign probability to the fraction c. Eq. (2) then turns $\Phi(a)$ into c. For small a, $\Phi(a) = 1/2 + a/\sqrt{2\pi} + O(a^3)$. The derivative of logit q at $q = 1/2$ is 4, so logit $\Phi(a) = 4a/\sqrt{2\pi} + O(a^3)$. At the final inverse temperature β_f, a field adds $\beta_f NGm$ to each configuration's log weight. Let $\bar{m}_f$ be the zero-field mean magnetization in the positive mode at β_f. To first order in G, the log probability of the positive mode changes by the average of this term over the mode, $\beta_f NG\bar{m}_f$, and that of the negative mode by the opposite amount. The equilibrium log-odds is thus approximately $2\beta_f NG\bar{m}_f$. Dividing the two leading log-odds gives

$$c \simeq \frac{\sqrt{2}\,\pi^{-1/4}}{\beta_f \bar{m}_f}(N^2\gamma)^{-1/4}.$$

Regime. The derivation needs β near 1 during selection, linear growth of m, and selection outside the critical window. Over the window, of duration $T \sim \gamma^{-1/2}$, the growth rate $\beta - 1 = \gamma t$ reaches $\gamma T \sim \sqrt{\gamma}$, small on a slow ramp ($\gamma \ll 1$), so the expansion near $\beta = 1$ holds. With a bounded, the other two conditions follow from $N\gamma \gg 1$. A larger N lowers the noise; faster cooling shortens the window, so less noise accumulates, and raises the growth rate, so the cubic term matters only at larger $|m|$. In scaling terms, the cubic term matters once $|m| \sim \gamma^{1/4}$; the noise accumulated over T, of order $\sqrt{T/N} \sim N^{-1/2}\gamma^{-1/4}$, is a fraction $(N\gamma)^{-1/2}$ of this scale, and the field's contribution, of order $|G|T$, a fraction $|G|\gamma^{-3/4} = \pi^{-1/4}a(N\gamma)^{-1/2}$, small for bounded a. Selection must also span more than the critical window, inside which the cubic term shapes the fluctuations of m and the Gaussian analysis above fails. To leading order in N, Stirling's formula and the interaction $N\beta m^2/2$ give the zero-field equilibrium density of m, $\log \pi_{\beta,0}(m) \simeq \text{constant} + N[(\beta-1)m^2/2 - m^4/12]$. At $\beta = 1$, fluctuations therefore reach $|m| \sim N^{-1/4}$, and the quadratic term matters when $|\beta - 1| \sim N^{-1/2}$: this range of β is the *critical window*. The β range crossed during selection, $\gamma T \sim \sqrt{\gamma}$, exceeds it when $N\gamma \gg 1$. These conditions allow linear selection before nonlinear growth fixes the magnitude; they do not exclude later switching.

Exact reduction and numerical evolution. To test the law, we compute the sampler's distribution exactly. Waiting times between CW updates are independent exponentials with mean $1/N$, so the count k of positive spins is a Markov chain whose rates, at a given temperature, depend only on k. Each spin updates once per expected sweep on average, so each rate is the number of eligible spins ($N - k$ negative or k positive) times their redraw probability:

$$W_k^+ = \frac{N-k}{2}\left[1 + \tanh\left(\beta\frac{2k - N + 1}{N-1} + h\right)\right], \tag{8}$$

$$W_k^- = \frac{k}{2}\left[1 - \tanh\left(\beta\frac{2k - N - 1}{N-1} + h\right)\right], \tag{9}$$

where $h = \beta G$; the $+1$ and -1 exclude the updated spin from the average of its neighbors. Equal probability flow between adjacent counts, $\pi_k W_k^+ = \pi_{k+1} W_{k+1}^-$ (*detailed balance*), holds for the distribution of k under Eq. (1),

$$\pi_k \propto \binom{N}{k} \exp\left[\frac{\beta}{2(N-1)}\big((2k-N)^2 - N\big) + h(2k - N)\right], \tag{10}$$

where the binomial factor counts configurations with the same k. With π_k at the final β, the equilibrium probability of Section 2 is $q_{\text{eq}} = \sum_{k>N/2}\pi_k / \sum_{k\neq N/2}\pi_k$; the omitted zero-magnetization

Table 2: Linear-ramp comparison with the selection law. Entries give exact c, c_{num}, and, in parentheses, its percentage excess over the law, $100(c_{\mathrm{num}}/c_{\mathrm{sel}} - 1)$.

N	$\gamma = 0.004$	$\gamma = 0.001$	$\gamma = 0.00025$
1024	0.09370 (0.93%)	0.13393 (1.81%)	0.18868 (1.03%)
2048	0.06587 (0.46%)	0.09372 (0.97%)	0.13314 (1.23%)
4096	0.04645 (0.23%)	0.06589 (0.49%)	0.09361 (0.86%)

mass is below 10^{-36} at $\beta_f = 1.6$ for $N \geq 512$. We evolve the count probabilities $P_k(t)$, $\dot{P}_k = W^+_{k-1}P_{k-1} + W^-_{k+1}P_{k+1} - (W^+_k + W^-_k)P_k$, by implicit Euler steps with Richardson extrapolation. The result has no sampling error, because the probabilities are evolved, not sampled.

Linear ramps. The ramps of Section 3 start in the equilibrium distribution at $\beta = 0.7$ and cool to $\beta = 1.6$, with $G = 0.0006125 \times 2048/N$; a fixed NG keeps the equilibrium log-odds nearly constant across sizes. In the evolution, β is piecewise constant at the midpoint of each interval $\Delta\beta = 0.005$, held for $0.005/\gamma$ expected sweeps, and each interval has $0.25/\gamma$ (rounded) Richardson-extrapolated implicit Euler steps of about 0.02. The discrete-update chain of Appendix B, computed without a time step, reproduces the $\gamma = 0.004$ column to the printed digits. Table 2 compares the resulting exact c, c_{num}, with the selection law c_{sel} of Section 3. The excess over c_{sel} about halves when N doubles at $\gamma = 0.004$ and 0.001, but not at 0.00025. We did not derive the correction that causes this excess.

Appendix B. Methods of the Hopfield test

This appendix describes the runs of Section 4, their predictions and their analysis, all fixed in a pre-registration before the runs.

Sampler. Each run redraws one uniformly chosen spin per update, independently of the other runs, from its conditional distribution under Eq. (5): $\Pr(x_i = +1 \mid x_{-i}) = [1 + \tanh(\beta h_i)]/2$, where x_{-i} denotes the other spins and $h_i = (N-1)^{-1} \sum_{j \neq i} \sum_\mu \xi^\mu_i \xi^\mu_j x_j + G\xi^1_i$, in double precision. A sweep is exactly N updates, and update $n = 0, \ldots, M-1$ uses $\beta_n = 0.7 + 0.9(n + \frac{1}{2})/M$, with $M = 0.9N/\gamma = 225N$, so the ramp lasts 225 sweeps. Runs start from independent unbiased spins, the equilibrium at $\beta = 0$; for the exact CW chain at $N = 2048$ this start and the equilibrium at $\beta = 0.7$ give the same c to six digits. Half the runs of each pattern set have field $+|G|$ and half $-|G|$, with $NG = 1.2544$ ($|G| = 0.0006125$ at $N = 2048$), so $a = 0.21$, 0.15 and 0.10 at $N = 1024$, 2048 and 4096.

One pattern is the CW chain. With $y_i = \xi^1_i x_i$, the local field satisfies $\xi^1_i h_i = (N-1)^{-1} \sum_{j \neq i} y_j + G$, the CW local field of y, so redrawing x_i redraws y_i from its CW conditional. Because the site is chosen independently of the state, the one-pattern sampler is the CW sampler of Eq. (1) at every finite N, for every pattern. Exact agreement needs random-site updates: permutation sweeps, which redraw every spin once per sweep in random order, define a different chain, which the count k alone does not describe.

Table 3: Hopfield runs. Pattern-1 runs enter the estimate; $m^1 = 0$ counts runs won by another pattern and excluded. Each row uses its own seeded PyTorch GPU generator.

p	N	runs	pattern-1 runs	$m^1 = 0$	GPU seed
1	1024	320,000	320,000	0	20271268
1	2048	640,000	640,000	0	20281508
1	4096	1,260,000	1,260,000	0	20301988
2	2048	640,000	321,598	4,318	20281509
3	1024	540,000	181,324	5,752	20271270
3	2048	820,000	274,222	6,619	20281510
3	4096	1,460,000	504,568	6,708	20301990
4	2048	940,000	254,495	7,780	20281511

Predictions. Each c_{CW} is computed for this discrete sampler: the distribution of k is multiplied by the transition matrix of each update in turn, with no time step. The two-pattern prediction propagates each block's distribution in the same way (Appendix C). For $p \geq 3$, $c_{\mathrm{pred}} = A_p c_{\mathrm{CW}}$ with A_p from numerical integration.

Patterns, labels and estimator. Pattern set $s = 0, \ldots, 49$ at size N consists of the first p of four rows of independent equiprobable signs drawn with NumPy `default_rng`$([20261027, N, s])$, so the sets are nested in p. At $\beta = 1.6$, each run is labeled by the pattern with the largest $|m^\mu|$ and the sign of m^1. Runs with a tie for the largest $|m^\mu|$ or with $m^1 = 0$ are excluded. No run ended in a tie, so $m^1 = 0$ occurred only in runs won by another pattern (Table 3), and no pattern-1 run was excluded. Pooled over pattern sets and field signs, $\widehat{q}$ is the fraction of pattern-1 runs whose m^1 has the field's sign, and $\widehat{c} = \operatorname{logit} \widehat{q} / \operatorname{logit} q_{\mathrm{eq}}^{\mathrm{CW}}(N)$, with the exact CW equilibrium log-odds at $\beta_f = 1.6$: 3.5742, 3.5747 and 3.5749 at $N = 1024$, 2048 and 4096. Pooling both signs cancels a sign bias of the sampler to first order. Intervals are percentile intervals of 10,000 cluster-bootstrap resamples of the 50 pattern sets (seed 7); a resampled set brings both signs. The size exponent of Section 4 is $\frac{1}{2} \log_2[\widehat{c}(1024)/\widehat{c}(4096)]$ at $p = 3$, with an interval from independent bootstraps of the two rows; the exact CW chain gives 0.506.

Equilibrium reference. By the map $y_i = \xi_i^1 x_i$, the Hopfield and CW equilibrium log-odds are equal at $p = 1$. At $p = 2$ the exact two-block calculation gives a Hopfield value 0.9993 times the CW one on average over our pattern sets; at $p = 3$, exact enumeration over the four sets of spins on which the patterns agree or differ gives ratios of 0.987–0.997 at $N = 512$ and 0.991–0.998 at $N = 1024$ (means 0.994 and 0.997). We did not compute it at $p = 4$.

Pre-registration. Before any run, we fixed the protocol, the predictions, the estimator and the run counts. The run counts aimed at a 95% half-width of at most 3% of the predicted c, under an assumed spread between pattern sets. Three gates had to pass before the $A_p c_{\mathrm{CW}}$ rows were read: G0, the GPU sampler reproduces a NumPy reference bit for bit on a given random stream ($N = 16$, $p = 3$, 50 updates, 400 runs), and the overlaps carried through every run equal recomputed ones; G1, each one-pattern interval contains c_{CW}; G2, the two-pattern interval contains the exact two-block value. A short pilot, never pooled with the results, led to two amendments before any row ran: all run counts were doubled, and the four gate rows were read at a joint 95% level, each with a 98.75%

interval. All gates passed. The pre-stated readings overlap for the $A_p c_{\mathrm{CW}}$ rows: every interval of $c/(A_p c_{\mathrm{CW}})$ contains 1 (reading a, which the protocol glossed as A_p holding to within 3%), but the deficits do not order with $p/(N\gamma)$, which the protocol read as a failure of independence or of the largest-amplitude rule (reading c, A_p not supported as stated). The differences between the deficits are smaller than their intervals, so these four rows cannot decide between the readings (Section 5). Table 1 reports 95% intervals for all rows. The runs used 4.1×10^{12} spin updates, about 2.9 GPU hours.

Appendix C. Competition among patterns

This appendix derives the amplification A_p of Eq. (6), the exact two-pattern prediction, and the condition under which pattern correlations leave A_p unchanged.

Conditioning on the winning pattern. Eq. (6) follows from conditioning on the pattern that wins. Under the assumptions of Section 4, let pattern 1's early amplitude (Appendix A) be $Y^1 \sim \mathcal{N}(\delta, \sigma^2)$, where δ is the field-induced mean shift and the other early amplitudes have zero mean and variance σ^2; Y^1 has the final sign of m^1, and δ/σ plays the role of a in Appendix A. The density f of Y^1 gives $f(r)/f(-r) = \exp(2r\delta/\sigma^2)$ at magnitude $r > 0$. Which pattern wins depends only on the magnitudes of the early amplitudes, so conditioning on it does not change these sign odds at fixed r:

$$\Pr(Y^1 > 0 \mid |Y^1| = r, \text{ pattern 1 wins}) = \frac{1}{1 + e^{-2r\delta/\sigma^2}} \simeq \frac{1}{2} + \frac{r\delta}{2\sigma^2}.$$

At $\delta = 0$, all patterns have the same distribution, so the winning magnitude is distributed as σM_p, where $M_p = \max_{1 \le \mu \le p} |Z_\mu|$ for independent standard Gaussians Z_μ. The distribution of the winning magnitude given that pattern 1 wins is even in δ, so its first-order change is zero. Averaging the sign probability over winning magnitudes gives, to first order, $q_{\mathrm{chain}} \simeq 1/2 + (\delta/2\sigma)\,\mathbb{E}M_p$, and expanding the logit around $1/2$ gives $\operatorname{logit} q_{\mathrm{chain}} \simeq (2\delta/\sigma)\,\mathbb{E}M_p$. With one pattern, $M_1 = |Z|$ and $q_{\mathrm{chain}} \simeq 1/2 + \delta/(\sigma\sqrt{2\pi})$, the first-order form of $\Phi(a)$. Dividing by this one-pattern response, with δ/σ unchanged by the other patterns and the same equilibrium log-odds, gives Eq. (6). The calculation averages sign probabilities before taking log-odds and gives sign odds given the winning pattern, not pattern frequencies.

Two patterns, exactly. In the variables $y_i = \xi_i^1 x_i$, the coupling of y_i and y_j in Eq. (5) is $(1 + \eta_i\eta_j)/(N-1)$, with $\eta_i = \xi_i^1\xi_i^2$: $2/(N-1)$ within the spins where the two patterns agree, $2/(N-1)$ within those where they differ, and zero between these sets, which therefore form non-interacting CW blocks, of 944–1104 spins in our pattern sets. With S_A the sum of the y_i where the patterns agree and S_B the sum where they differ, $Nm^1 = S_A + S_B$ and $Nm^2 = S_A - S_B$, so pattern 1 wins exactly when S_A and S_B have the same sign, and its sign is then that of S_A. If the blocks evolve independently, the odds of a positive sign given that pattern 1 wins are the product of the blocks' odds, so their logits add. Under random-site updates, a block is chosen with probability equal to its share of the spins; we propagate each block's count distribution exactly and treat the blocks as independent, which they are up to the shared number of updates (a brute-force check at $N = 10$ agrees to 10^{-5} relative). Pooling over pattern sets as the estimator does gives $c_{\mathrm{pred}} = 0.09363$ at $N = 2048$; the per-set values span 0.09360–0.09400. By Eq. (3), each block's sampled log-odds grow as the square root of its size, so two blocks of $N/2$ spins give $\sqrt{2}$ times the log-odds of one block of N: the selection law reproduces $A_2 = \sqrt{2}$, and $A_2 c_{\mathrm{CW}} = 0.09316$ lies 0.5% below the exact value.

When correlations matter. For p patterns, $C_{\mu\nu} = N^{-1}\sum_i \xi_i^\mu \xi_i^\nu$ records their pairwise correlations. At zero field, averaging the expected Gibbs redraw over sites, weighted by ξ_i^μ, gives to leading order in $1/N$

$$\dot{m}^\mu \simeq -m^\mu + \frac{1}{N}\sum_i \xi_i^\mu \tanh\left(\beta \sum_\nu \xi_i^\nu m^\nu\right) \simeq -m^\mu + \beta \sum_\nu C_{\mu\nu} m^\nu,$$

using $\tanh z \simeq z$ for small overlaps. The drift of $\boldsymbol{m} = (m^1, \ldots, m^p)$ is thus $(\beta C - I)\boldsymbol{m}$. The component of $\boldsymbol{m}$ along an eigenvector of C with eigenvalue λ grows at rate $\beta\lambda - 1$, beginning at $\beta = 1/\lambda$ rather than at $\beta = 1$ as for $C = I$. Each off-diagonal entry averages N independent random signs, so its standard deviation is $N^{-1/2}$, and the eigenvalues spread over a range of order $(p/N)^{1/2}$. On a linear ramp, the selection window spans a β range of order $\sqrt{\gamma}$ (Appendix A), so the components start growing at nearly the same point of the window only when $(p/N)^{1/2} \ll \sqrt{\gamma}$, that is, $N\gamma \gg p$. Otherwise the amplitudes have unequal variances and growth factors, and the assumptions behind Eq. (6) fail. Before the runs, a linear Gaussian model of these components, which we did not validate, gave expected deficits of 0.5–3% for the four $A_p c_{\mathrm{CW}}$ rows; the measured deficits are 0.5–6%.